SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section, where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	3AM
Project / Solution Name	Pathwaay
Name (Leader)	Ooi Wei Lun
Email & Phone (Leader)	lunweii1025@gmail.com & 0197107401
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU)
Live Prototype URL	https://3-am-sdc-hackathon.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/weiiilunnnn/3am-sdc-hackathon

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Pathwaay combines a Next.js/React/TypeScript interface with Python FastAPI APIs. Registration, fictional CV review and profile skills feed a local SQLite database. An official pinned ESCO v1.2.1 index supplies occupation/skill relationships, a deterministic scorer compares recognised profile skills with documented role requirements and generates skill-gap views. Reskilling uses reviewed provider links, optional Tavily Search/Extract discovers candidate learning pages with content checks. OpenAlex publication counts and attributed Remote OK/Himalayas posting feeds provide limited research/job evidence, separately labelled and cached. The separate Pathwaay Explorer uses locally self-trained scikit-learn Random Forest models on synthetic data for illustrative role/scenario outputs. Optional RunPod Serverless GLM-5.3 supports source-constrained explanations and reviewed skill extraction, calls require backend-only credentials and can fail without corrupting the original evidence. The Career Simulator saves a user-selected ESCO exploration path and life milestones to SQLite using demo-only browser capability scoping. The project is run locally or through the Replit development launcher, production hosting, authentication and durable cloud storage require verification.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Next.js 16.4.2	Web pages, routing and API proxy	MIT
React / React DOM 19.3.1	Interactive frontend components	MIT
TypeScript 5.9.3	Typed browser and UI code	Apache-2.0
Python 3.12	Backend, data processing and modelling	PSF License
FastAPI >=0.115,<1.0	Validated Python HTTP API	MIT
Uvicorn >=0.30,<1.0	Development/API ASGI server	BSD-3-Clause
Pydantic (installed v2 series)	Schema and data validation	MIT
scikit-learn >=1.5,<2.0	Self-trained Random Forest models	BSD-3-Clause
joblib >=1.4,<2.0	Save/load local trained models	BSD-3-Clause
RunPod SDK >=1.7,<2.0	Optional RunPod client dependency	MIT
pypdf	Fictional PDF CV text extraction	BSD-3-Clause
openpyxl >=3.1,<4.0	Spreadsheet handling	MIT

Name & Version	Purpose in Project	License
python-dotenv >=1,<2	Backend environment loading	BSD-3-Clause
SQLite (embedded)	Persistent demo profiles and caches	Public domain
Playwright 1.64.1	Browser tests / validation	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless API (GLM-5.3)	Optional AI narrative, CV/skill extraction, career explanations and insights	API key, event access reported
RunPod Serverless (RF endpoint)	Optional remote synthetic-career inference, local RF fallback	Configured endpoint only
Tavily Search API	Optional discovery of reskilling provider URLs	Key/credits required
Tavily Extract API	Read candidate course content for validation	Key/credits required
ESCO official web service (EC)	Built local 2026 ESCO occupation/skill snapshot, optional manual lookup	Public EC API / reuse conditions
OpenAlex Works API	Refresh indexed research-activity observations	Public API, CC0 data, optional key
Remote OK public feed	Attributed remote-job snapshot, manual cached refresh	Public feed, conditional terms/credit
Himalayas jobs API	Optional job-posting signal adapter	Public API, use/reuse terms verify

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development/demo launcher, app hosting capability	Plan, region and public deployment
RunPod Serverless	Optional remote language-model / career-model inference	Endpoint model, GPU region
Local development machine	Next.js frontend + FastAPI backend	Not cloud hosting, local demonstration
SQLite database	Profile, cached sources, simulator journey, job state	Local file, no managed database service
Authentication	No external auth/managed identity integrated	None, prototype-only local demo controls

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GLM-5.3	RunPod-hosted LLM, underlying host/weights not independently attested	Pre-trained, optional generated explanations / skill extraction	Hosted provider/model terms
RandomForestClassifier, synthetic-random-forest-v1	Project / scikit-learn	Self-trained on synthetic career-profile CSV	Project-created weights, BSD-3 code, data provenance
RandomForestRegressor, scenario model v1	Project / scikit-learn	Self-trained on synthetic skill-scenario CSV	Project-created weights, BSD-3 code, data provenance
Deterministic ESCO relevance	Project / official ESCO	Rule-based weighted skill overlap, no learned model	ESCO EC reuse, project methodology

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO v1.2.1 (3,658 occupations, 14,579 skills)	EC official ESCO API / esco.ec.europa.eu	Public	EC reuse conditions 2011/833/EU	N
OpenAlex Works research-year observations	api.openalex.org/works	Public	CC0 1.0	N
Remote OK posting snapshot	remoteok.com/api	Public feed	Conditional reuse/attribution	N
Himalayas job feed (optional)	himalayas.app/jobs/api	Public feed	Provider terms / verify	N
unified_profile_career_dataset.csv	Repository /data	Synthetic	Team-created	N
synthetic_skill_scenarios.csv	Repository /data	Synthetic	Team-created	N
synthetic_sdg_skill_mappings.csv	Repository /data	Synthetic	Team-created	N
sdg_goal_reference.csv	Repository /data (UN target summary)	Curated public reference	Attribution/reuse verify	N
Reviewed course/resource catalogue	Repository /data/resources, provider URLs	Curated links	Provider terms vary, link only	N
Fictional demo profiles / company	Repository /data/demo and app sample profiles	Synthetic	Team-created	N

* Public job listings may contain names/contact details upstream, only bounded metadata is stored, while descriptions are processed transiently. Rights and sources: repository docs/DATA_SOURCES_AND_LICENSES.md and docs/employment-source.md (main, 24 Oct 2026).

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score, undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex	Code generation/refactoring, test creation, debugging and validation	Moderate
ChatGPT	Architecture planning, drafting project handoffs and submission text	Major
Microsoft Copilot	Code generation/refactoring, test creation, debugging and validation	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Team members review AI-assisted changes against source data, code structure and intended features. Backend tests, frontend component/logic tests, TypeScript checks, production builds and selected browser flows were reported, including checks for ESCO provenance, path persistence and provider-error handling. Generated career explanations are validated against known ESCO identifiers and citations, unverifiable claims are rejected or labelled as optional interpretations. Synthetic model outputs are labelled illustrative, not real forecasts. Human reviewers remain responsible for correctness, security, licensing and final submission. All AI-assisted edits must be rechecked on the final main commit before release.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	Event-provided inference access reportedly used	Configured GLM explanations/skill extraction, verify account credit, sponsor and actual use
Replit	Development platform / possible event access	Repository contains Replit proof-of-use artifact, confirm credits and hosted URL

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod enables optional hosted language-model features such as explanations and skill extraction, it is not necessary for locally sourced ESCO relevance, reviewed learning links, the Career Simulator or the synthetic Random Forest baseline. Tavily-enhanced course discovery is similarly optional and falls back to local reviewed resources. Replit can launch a development demonstration, but equivalent local hosting is possible. Before signing, the team must confirm whether RunPod/Replit access or credits were officially sponsor-provided, their provider account conditions, and the precise scope of actual usage.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application is an unauthenticated prototype intended for fictional profiles and sample CVs only, it must not be used with genuine personal or employee records. Registration may hold profile text, skill selections and an optional email in browser session storage and SQLite, the CV file is parsed in memory and is not retained as an upload. Optional RunPod requests send bounded fictional text or selected career/skill context after user action, full names, emails and raw CV files are excluded from designated skill-analysis flows, although certain chapter notes can be transmitted on explicit request. RunPod provider retention is not verified. External research and job sources are cached with provenance. API keys remain backend-side in environment variables. SQLite data has no automatic retention deadline, users should delete demo data manually. Browser-held profile references and simulator capabilities are not secure account authentication.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

ESCO is a European skills classification, weighted overlap does not prove qualification, transition likelihood or Malaysian hiring demand. OpenAlex counts represent research activity, Remote OK/Himalayas samples are incomplete job-board snapshots. Synthetic Random Forest career and scenario outputs are illustrations, not validated forecasts. Hosted GLM text may hallucinate or violate source-grounding rules: the backend constrains evidence IDs, checks outputs and withholds invalid interpretations. RunPod and Tavily can rate-limit, time out or require paid credits, caching, bounds, optional requests and deterministic fallbacks limit disruptions. Course coverage, prices and prerequisites can remain unverified and are labelled accordingly. Local browser-capability storage lacks verified identity, recovery and cross-device access. Public deployment needs authentication, retention policies, provider-term review and end-to-end verification, especially for third-party keys and sponsor resources.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Date: 25-10-2026